

Prof. Puneet Azad

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Dr. Puneet Azad is working in this Institute since 2002. He fulfills all his duties of teaching, administration and research with keen interest. He has taught several courses to B.Tech students such as Communication systems, Digital Communication, Electronic devices, Digital Electronics, Signals and Systems, Digital signal processing, Wireless Communication, Analog Electronics, Telecommunication Networks. In research, he has explored different electronic materials for a variety of applications in low power electronics. His main contribution is towards designing electronic circuits for power optimization suitable for sensors. Energy-harvesting techniques such as triboelectricity, pyroelectricity, piezoelectricity and thermoelectricity have been used to harvest energy. His group is actively involved to sustainable energy solutions with main focus on materials engineering and designing energy efficient circuits. These technologies will be playing important role towards make in India mission and these devices are next generation technological solutions. His aim is to demonstrate proof of concepts based on above mentioned concepts. Various prototypes have been fabricated and reported in various journals of international repute.

Dr. Puneet has authored and published **22 research** papers in SCI and Scopus indexed journals and contributed his work in thirteen conferences. He is currently supervising three students registered at GGSIPU University, Delhi. He has been awarded **Best Faculty Award** outstanding performance in academics and administration in MSIT in the year 2014. He also received **Research Excellence Certificate** in the year 2018 and 2019 for his contribution in research.

He has worked at several position in MSIT including NAAC/IQAC Coordinator, NBA Coordinator, Dean Infrastructure, Dean Academics, GGSIPU Approval / AICTE affiliations Convener, Convener/Team Member in various Conferences, Audits, Head of Department, Convener of Alumni Association, Teacher representative, Convener of Technical festival etc

EDUCATION AND DEGREES AWARDED

- **Ph.D.** in “Energy Efficient Clustering Algorithms for Wireless Sensor Networks” from **Gautam Buddha University**, Greater Noida in 2014.
- **Master of Engineering (M.E.)** in Electronics and Communication from **Delhi College of Engineering**, Delhi in 2000.

- **B.E.** in Electronics from **Bhilai Institute of Technology** (Pt. Ravi Shankar Shukla University, Raipur), in 1999

AWARDS AND HONORS

- **Guest Editor Special Issue "Ferroelectric Materials for Energy Harvesting" of Crystals** (https://www.mdpi.com/journal/crystals/special_issues/Energy_Harvest)
- **Award of INSA Visiting Scientist Programme 2020-21, 2022-23**
- **Research Excellence Certificate** in recognition of an outstanding contribution to the quality of research in 2018 and 2019 in MSIT
- **Best Faculty Award 2014** for outstanding performance in academics and administration in MSIT
- **Graduate Aptitude Test of Engineering** (Gate' 99) with 94.7 percentile in 1999
- **Microsoft Certification** in desktop application in VC++ (MCSD)
- **Merit Certificate** in Mathematics in XII standard in 1994

KEY PUBLICATIONS

SCI Indexed Journals

1. **Puneet Azad**, Deepshikha Yadav, Aayush Kumar, Rakhi Kamra, Koyel Datta Gupta, Archana Balyan, "Impact of energy efficient green data centers: structures and methodologies systematic review," Proceedings on Engineering Sciences, Vol. 07, pp. 1749-1758, 2025
2. D Nath, **P. Azad**, S.M. Ibrahim, V.P. Singh, Antibacterial and photocatalytic performance of eggshell-derived CaF₂-mortar cement composites, Journal of the Australian Ceramic Society, Vol. 59, pp.93-103, 2023 (IF: 2.6)
3. Saurav Sharma, Raj Kiran, **Puneet Azad**, Rahul Vaish, A review of piezoelectric energy harvesting tiles: Available designs and future perspective, Energy Conversion and Management (Elsevier), vol. 254, 115272, 2022 (IF: 11.8)
4. **Puneet Azad**, Saptarshi Karmakar, Anuruddh Kumar, Sobhy M. Ibrahim & Rahul Vaish, An optimization study on (Ba_{0.85}Ca_{0.15})(Zr_{0.1}Ti_{0.9})O₃-based piezoelectric energy-harvester using finite element method, Journal of the Australian Ceramic Society (Springer), <https://doi.org/10.1007/s41779-021-00690-2>, 2021 (IF: 2.6)
5. Imen Kebaili, HI Elsaedy, Imed Boukhris, Lamia Ben Farhat, Samia Ben Ahmed, ZA Alrowaili, MS Al-Buriah, **Puneet Azad**, Poling direction-driven tuning of piezoelectric properties of ferroelectric materials: an experimental study, Journal of the Australian Ceramic Society, Vol. 58, pp. 1111-1115, 2022 (IF: 2.6)
6. **Puneet Azad**, Temperature controlled voltage regulated boost converter for thermoelectric energy harvesting, IETE Journal of Research (Taylor & Francis), doi:10.1080/03772063.2019.1651678, 2019 (IF: 1.4)
7. Nada Alfryyan, Sandeep Kumar, Samia Ben Ahmed, Imen Kebaili, Imed Boukhris, **Puneet Azad**, Mohammed Sultan Al-Buriah, Rahul Vaish, Electric poling effect on piezocatalytic BaTiO₃/polymer composites for coatings, Catalysts, Vol.12, 1228, 2022 (IF: 4.5)
8. Parul Chaudhary and **Puneet Azad**, Energy harvesting from human biomechanical energy for health monitoring devices, IETE Journal of Research (Taylor & Francis), vol.67, pp. 74-81, 2021 (IF: 1.4)
9. Deepshikha Yadav, **Puneet Azad** and Rahul Vaish, Solar energy harvesting using candle-soot-

- coated thermoelectric materials, *Global Challenges* (Wiley), vol.4, 1900080, 2020 (IF: 7.4)
10. Parul Chaudhary, **Puneet Azad**, Energy harvesting using shoe embedded with piezoelectric material, *Journal of Electronic Materials* (Springer), vol. 49, pp. 6455-64, August 2020 (IF: 2.8)
 11. Khushboo and **Puneet Azad**, Design and analysis of a synchronized interface circuit for triboelectric energy harvesting, *Journal of Electronic Materials* (Springer), vol. 49, pp. 2491–2501, 2020 (IF: 2.8)
 12. **Puneet Azad**, Samriddhi Raut and Rahul Vaish Candle soot coated egg carton material for oil water separation and detergent adsorption, *Bulletin of Materials Science* (Indian Academy of Sciences), vol.43, Article-7, 2020 (IF: 2.1)
 13. Anuruddh Kumar, Sidhant Kumar, Satyanarayan Patel, Moolchand Sharma, **Puneet Azad**, Rahul Vaish, Rajeev Kumar, K.S. Srikanth, Pyroelectric energy conversion harvesting using Ba_{0.85}Sr_{0.15}Zr_{0.1}Ti_{0.9}O₃ ceramics and its cement based composites, *Journal of Intelligent Material Systems and Structures* (SAGE Publications), vol. 30, pp.869-877, 2019 (IF: 2.7)
 14. **Puneet Azad**, Khushboo and Rahul Vaish, Solar energy harvesting using pyroelectric effect associated with piezoelectric buzzer, *Physica Status Solidi (a)* (Wiley), vol. 216, doi:10.1002/pssa.201900440, 2019 (IF: 2.0)
 15. **Puneet Azad**, Moolchand Sharma and Rahul Vaish, Diesel exhaust emission coated pyroelectric materials for improved thermal energy harvesting, *Global Challenges* (Wiley), vol.3, 1800089, 2019 (IF:7.4)
 16. Khushboo and **Puneet Azad**, Triboelectric energy harvester using human biomechanical motion for low power electronics, *Bulletin of Materials Science* (Indian Academy of Sciences), vol. 42, <https://doi.org/10.1007/s12034-019-1801-9>, 2019 (IF: 2.1)
 17. Deepshikha Yadav and **Puneet Azad**, Low-cost triboelectric sensor for speed measurement and weight estimation of vehicles, *IET Intelligent Transport Systems*, vol. 12, pp. 958-64, 2018 (IF: 2.7)
 18. **Puneet Azad**, V.P. Singh and Rahul Vaish, Candle soot-driven performance enhancement in pyroelectric energy conversion, *Journal of Electronic Materials* (Springer), vol. 47, pp. 4721-30, 2018 (IF: 2.8)
 19. Moolchand Sharma, V.P. Singh, Shatrughan Singh, **Puneet Azad**, Bouraoui Ilahi and Niyaz Ahamad Madhar, Porous Ba_{0.85}Ca_{0.15}Zr_{0.1}Ti_{0.9}O₃ ceramics for pyroelectric applications, *Journal of Electronic Materials* (Springer), vol. 47, pp. 4882-91, 2018 (IF: 2.8)
 20. **Puneet Azad** and Rahul Vaish, Portable triboelectric based wind energy harvester for low power application, *European Physical Journal Plus* (Springer), vol. 132, pp.253-259, 2017 (IF: 2.9)
 21. **Puneet Azad** and Vidushi Sharma, Pareto optimal clustering scheme using data aggregation for wireless sensor networks, *International Journal of Electronics* (Taylor & Francis), vol. 102, No.7, pp.1165-1176, 2015 (IF: 1.2)
 22. **Puneet Azad** and Vidushi Sharma, “Energy efficient clustered scheme for Wireless sensor networks using multi-criteria decision making approach,”*International Journal of Computer Aided Engineering and Technology*, Inderscience, vol. 6, No.3, pp.324-336, 2014

PAPERS IN INTERNATIONAL CONFERENCES

1. Puneet Azad, Piezoelectric energy harvesting from pavements: A State-of-the-Art Review,

NCRTCES, MSIT, New Delhi, 2024

2. Puneet Azad, Power optimization in Triboelectric Energy Harvester, Elsevier Int. Elect. Electro. Engg. Conf. (IEEEEC-2021), DTU, 23-25 October, 2021
3. Parul Chaudhary and Puneet Azad, Efficiency Enhancement in a Piezoelectric Energy Harvester for a Low-Frequency System Using Sync. Charge Inversion, Innovations in Cyber Physical Systems, Lect. Notes Elect. Engg., pp 497-503, 2021
4. **Puneet Azad** and Rahul Vaish, "Large energy harvesting using simultaneous application of pyroelectric and thermoelectric materials," **IEEE Joint ISAF-ICE-EMF-IWPM-PFM conference - 2019 f2cπ2**, Lausanne, Switzerland, 14-19, 2019
5. Deepshikha Yadav and **Puneet Azad**, "Experimental analysis of power generation for ultra-low power wireless sensor nodes using various coatings on thermoelectric energy harvester," **IEEE 6th International Conference on Signal Processing and Integrated Networks (SPIN)**, Amity University, Noida, Mar 7-8, 2019
6. Khushboo and **Puneet Azad**, "Comparison of two design methods of triboelectric nanogenerator for building efficient energy harvesting and storage techniques," **Int. Conf. Manufacturing, Advance Computing, Renewable Energy and Communication (MARC 2018)**, Springer, HMR Institute of Technology and Management, July 19-20, 2018
7. Khushboo and **Puneet Azad**, "Design and implementation of conductor to dielectric lateral sliding TENG mode for low power electronics," **Int. Conf. of Signals, Machines and Automation (SIGMA 2018)**, Springer, Netaji Subhash Institute of Technology, New Delhi, pp. 1499-1502, February 23-25, 2018.
8. Deepshikha Yadav and **Puneet Azad**, "Design and implementation of robust low cost and low power prototype for generic counting system" **IEEE International Conference on Computing Communication and Automation (ICCCA2017)**, Galgotias University, G. Noida, pp. 1493-1498, May 5-6, 2017
9. Parul Chaudhary and **Puneet Azad**, "Demonstration of double electrode vertical-sliding triboelectric generator" **IEEE International Conference on Computing Communication and Automation (ICCCA2017)**, Galgotias University, pp. 1483-1487, May 5-6, 2017
10. Khushboo and **Puneet Azad**, "Triboelectric Nanogenerator based on Vertical Contact Separation Mode for Energy Harvesting" **IEEE International Conference on Computing Communication and Automation (ICCCA2017)**, Galgotias University, May 5-6, 2017
11. **Puneet Azad**, "Single and double electrode based Nanogenerators for energy harvesting from different materials" **IEEE International conference on Computing for Sustainable Global Development (INDIACom)**, Bharati Vidyapeeth's College of Engineering, New Delhi, March 1-3, 2017
12. **Puneet Azad**, "Energy harvesting from different materials using mechanical motion" **RSC-NPL Symposium on Advanced Nanomaterials for Energy**, October 7, 2016
13. **Puneet Azad** and Vidushi Sharma "Clusterhead selection using multiple attribute decision making (MADM) approach in wireless sensor networks", **QSHINE 2013 - 9th International Conference on Heterogeneous Networking for Quality, Reliability, Security and Robustness**, Springer, Gautam Buddha University, pp. 141-154, 11-12 January, 2013.

PATENTS PUBLISHED

1. **IoT BASED SMART COMPLAINT BOX** (Patent filed on 15/01/2021 & Published on

22/01/2021, Application No: 202111001925, The Patent Office Journal No. 04/2021 dated 22/01/2021, Part 1 & Page No: 3103)

ARTICLES PUBLISHED ON COVER PAGE OF JOURNALS



Khushboo and **Puneet Azad**, “Triboelectric energy harvester using human biomechanical motion for low power electronics,” **Bulletin of Materials Science (Indian Academy of Sciences)**, vol. 42, <https://doi.org/10.1007/s12034-019-1801-9>, 2019

BOOK CHAPTERS

- **Puneet Azad** and Vidushi Sharma, “**Clustering for Energy Efficiency in Wireless Sensor Networks**” Energy-Efficient Wireless Sensor Networks, CRC Press, Taylor & Francis Group, 2017
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